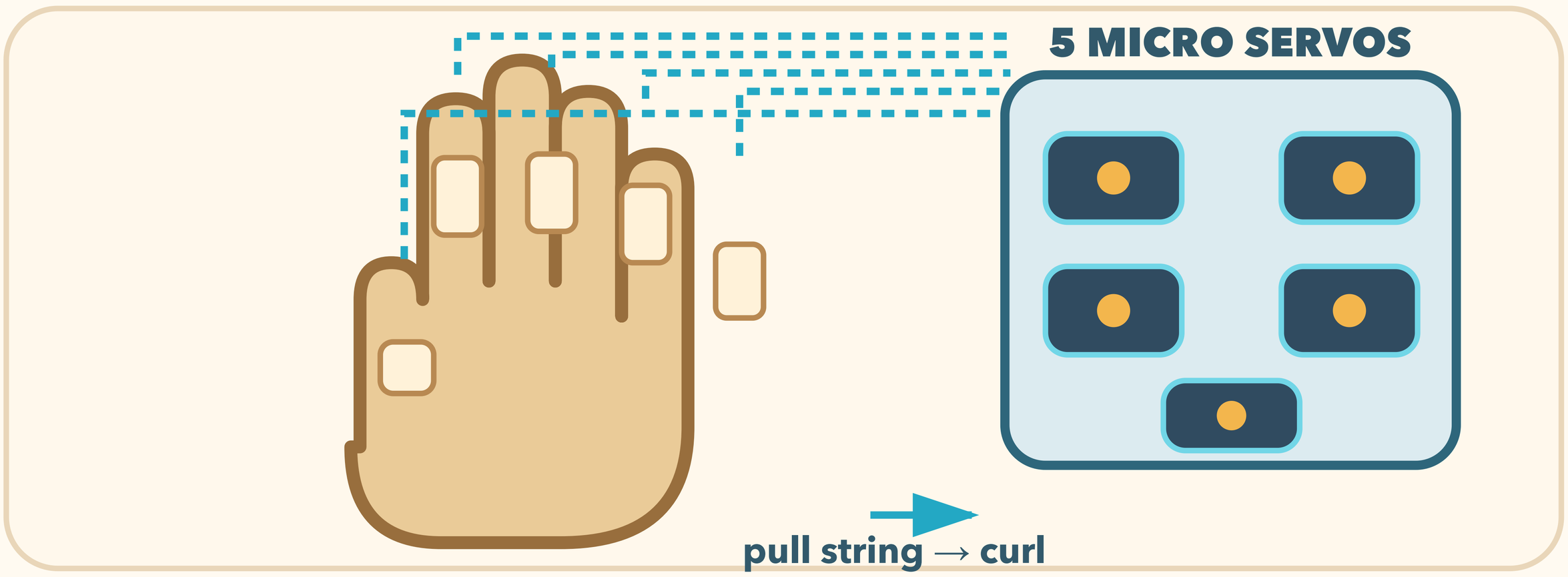
MAKE THE FINGERS MOVE

BUILD A CARDBOARD ROBOT HAND.

Five small servos pull strings to copy your tracked finger curl.



Gather these parts

- ✓ Arduino Uno or Nano
- ✓ Cardboard for palm + fingers
- ✓ USB data cable
- ✓ Rubber bands or elastic
- ✓ Computer + webcam
- ✓ 5 micro servos
- ✓ 5V, 2–3A external supply
- ✓ Thread or thin string
- ✓ Tape, glue, scissors
- ✓ Jumper wires

Build in 4 steps

1 Cut the hand

Trace a palm and five cardboard fingers. Add tape hinges at each finger joint so they can bend.

2 Thread the fingers

Tape a string near each fingertip and route it behind the finger to a servo horn. Use elastic to reopen it.

3 Mount the servos

Secure five servos to a cardboard wrist/base. Start with loose strings; tighten only a little at a time.

4 Test gently

Keep the hand clear, then move one finger slowly. Adjust the string before trying the full hand.

Wire five servos

FINGER	SIGNAL WIRE
Thumb	D3
Index	D5
Middle	D6
Ring	D9
Pinky	D10

Power matters: all servo red wires → external +5V. All brown/black wires → supply GND. Connect Arduino GND to that same GND. Keep the Arduino on USB for data. **Do not power five servos from the Arduino 5V pin.**

```
Servo signal colors are often:
yellow/orange = signal · red = +5V
brown/black = GND
Check your servo label/wire guide.
```

Load the project code

1. In Arduino IDE, open `arduino/servo_controller.ino`, choose Uno/Nano + port, then upload.

2. In `python_pipeline`, install packages:

```
python3 -m pip install -r requirements.txt
```

3. Start the webcam tracker:

```
python3 tracker.py
```

4. Press `s` in the video window to connect to the Arduino. Press `q` to quit.

What the maker build does

Python estimates five finger-curl values from the webcam. The serial bridge sends them to the Arduino. The Arduino smooths the values and moves five servos.

This build copies finger motion only. It does not include the browser twin's arm or gift pickup, and the physical Python path does not have the browser demo's gesture safety filter.

Build safely with an adult

Unplug servo power before changing wires or strings. Keep fingers clear of pinches. Use the external 5V supply, check servo travel, and stop if anything gets hot or strains. Adult help for cutting and wiring.

Get the code

Type this address to open the project files:

github.com/kcmagik-blip/cybergrip-lab